TEMASEK
JUNIOR COLLEGE

CHEMISTRY 8873/01

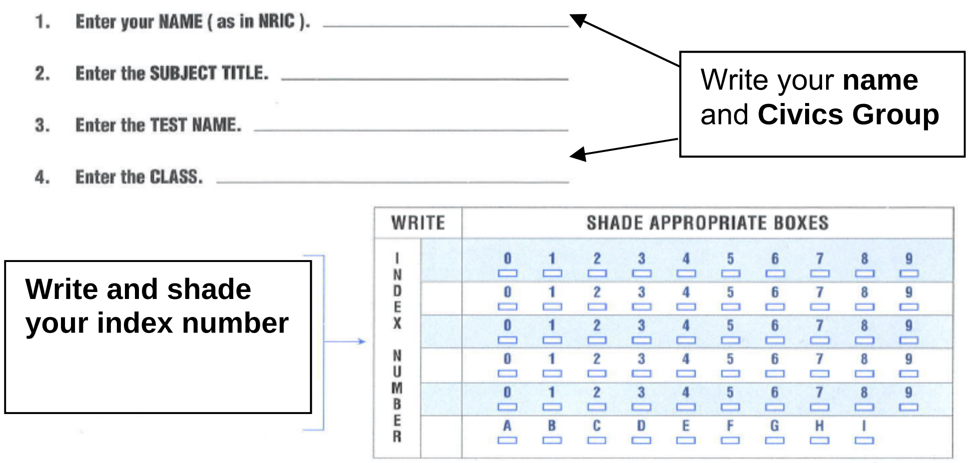
Paper 1 Multiple Choice

17th September 2018
1 hour

Additional materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.



There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **13** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate Answer Sheet (OMS).

1 Some isotopes are unstable and undergo nuclear (radioactive) reactions. In one type of reaction, an unstable nucleus assimilates an electron from an inner orbital of its electron cloud. The net effect is the conversion of a proton and an electron into a neutron.

Which of the following describes this type of reaction?

A $^{11}C \rightarrow {}^{12}C$

B $^{111}I \rightarrow {}^{111}Te$

C $^{76}Br \rightarrow {}^{75}Br$

D $^{76}Kr \rightarrow {}^{75}Br$

2 Sodium hydrogencarbonate can be prepared from sodium sulfate by a three-step process:

$Na_2SO_4(s) + 4C(s) \rightarrow Na_2S(s) + 4CO(g)$

$Na_2S(s) + CaCO_3(s) \rightarrow CaS(s) + Na_2CO_3(s)$

$Na_2CO_3(s) + H_2O(l) + CO_2(g) \rightarrow 2NaHCO_3(s)$

What is the mass of sodium hydrogencarbonate that could be formed from 100 kg of the sodium sulfate, assuming a 90% yield in each step?

A 106 kg **B** 96 kg **C** 86 kg **D** 43 kg

3 Which of the following reactions is a redox reaction?

A $2NO_2 \rightarrow N_2O_4$

B $6HCl + As_2O_3 \rightarrow 2AsCl_3 + 3H_2O$

C $SbF_3 + F_2 \rightarrow SbF_5$

D $Cr_2O_7^{2-} + 2OH^- \rightleftharpoons 2CrO_4^{2-} + H_2O$

4 When 2.6 g of a metal X are added to copper(II) sulfate solution 4.8 g of copper are obtained. The relative atomic mass of X is 52. Which one of the following cations of X is produced?

A X^{4+} **B** X^{3+} **C** X^{2+} **D** X^{+}

5 20.0 cm^3 of 0.02 mol dm^{-3} bromate(V), BrO_3^-, was found to react completely with 80.0 cm^3 of 0.01 mol dm^{-3} hydroxylamine, NH_2OH. BrO_3^- ions are reduced as follows:

$$BrO_3^- + 6e^- + 6H^+ \rightarrow Br^- + 3H_2O$$

Which of the following could be the half-equation for the oxidation of hydroxylamine?

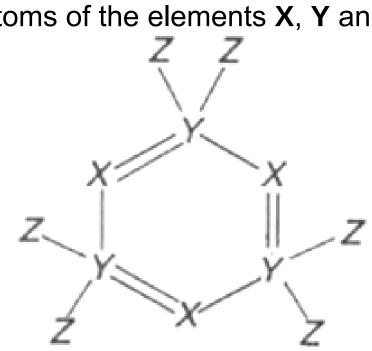
A $NH_2OH \rightarrow \frac{1}{2} N_2O + 2H^+ + \frac{1}{2} H_2O + 2e^-$

B $NH_2OH + H_2O \rightarrow NO_2^- + 5H^+ + 4e^-$

C $NH_2OH \rightarrow NO + 3H^+ + 3e^-$

D $NH_2OH + 2H_2O \rightarrow NO_3^- + 7H^+ + 6e^-$

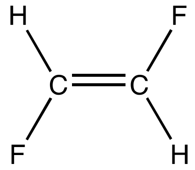
6 A stable molecule containing atoms of the elements **X**, **Y** and **Z** has the following structure.

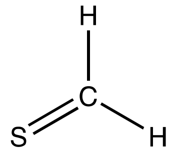


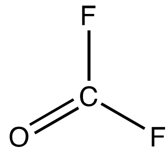
Which one of the following is a possible combination of elements?

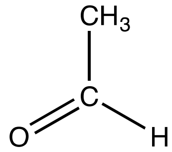
	X	**Y**	**Z**
A	P	As	H
B	P	N	C*l*
C	B	C	H
D	O	N	F

7 Which molecule has the largest dipole?

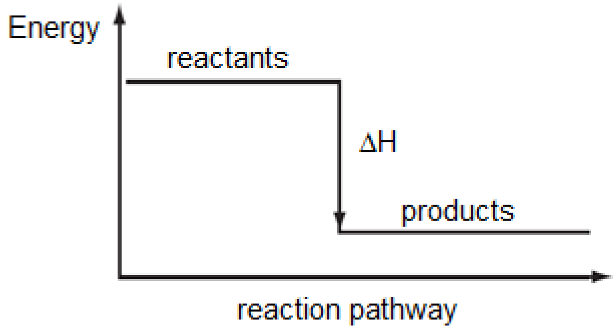
A 

B 

C 

D 

8 Which enthalpy change could **not** be correctly represented by the enthalpy diagram shown?



A Bond energy

B Lattice energy

C Standard enthalpy change of combustion

D Standard enthalpy change of neutralisation

9 Phosphorus pentachloride reacts with limited amount of water to give a liquid and white fumes as shown in the equation below.

$$PCl_5(s) + H_2O(l) \rightarrow POCl_3(l) + 2HCl(g) \qquad \Delta H^{\ominus} = -640 \text{ kJ mol}^{-1}$$

The following enthalpy changes are given:

$\Delta H_f^{\ominus}$ $PCl_5(s)$ $= -444$ kJ mol^{-1}
$\Delta H_f^{\ominus}$ $HCl(g)$ $= -92$ kJ mol^{-1}
$\Delta H_c^{\ominus}$ $H_2(g)$ $= -286$ kJ mol^{-1}

What is the standard enthalpy change of formation of $POCl_3$(l)?

A – 1278 kJ mol^{-1} **B** – 1186 kJ mol^{-1} **C** + 94 kJ mol^{-1} **D** + 274 kJ mol^{-1}

10 Compound **M**, $CH_3CH{=}CHCHClCH_3$, reacts readily with alcoholic KCN according to the following equation:

$$CH_3CH{=}CHCHClCH_3 + CN^- \rightarrow CH_3CH{=}CHCH(CN)CH_3 + Cl^-$$

The following kinetics data were collected:

Experiment	**[M] / mol dm^{-3}**	**[CN$^-$] / mol dm^{-3}**	**Relative Rate**
1	0.1	0.1	1
2	0.2	0.1	2
3	0.3	0.3	3

Which conclusion can be drawn from this information?

A The overall order of reaction is 2.

B Increasing the concentration of CN^- ions leads to the increase in the rate of reaction.

C The reaction is an addition reaction as it involves the two reactant molecules combining to form the product.

D Replacing the chlorine atom in **M** with bromine atom will require a shorter time for the reaction to be completed

11 The radioactive decay of isotopes **P** and **Q** follow first-order kinetics.

Isotope **P** decreases from 1800 counts per minute to 450 counts per minute in six months. Isotope **Q** decreases from 5400 counts per minute to 1350 counts per minute in four months.

In a separate experiment, a sample containing a mixture of the two isotopes was left to decay, and the molar ratio of **P** : **Q** was found to be 1 : 1 after six months.

What is the molar ratio of **P** : **Q** at the beginning?

A 1 : 2

B 1 : 4

C 2 : 1

D 4 : 1

12 Which of the following statements are correct?

1 From silicon to sulfur, the melting points of phosphorous and silicon are the lowest and highest respectively.

2 The maximum oxidation number of the element in the oxides increase continuously from sodium to chlorine.

3 The first ionisation energies of the elements in Period 3 increase continuously from sodium to chlorine.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

13 **X**, **Y** and **Z** are elements in Period 3 of the Periodic Table. The following statements were made about the properties of **X**, **Y** and **Z**, and their compounds.

1 The oxide of **Z** does not dissolve in excess dilute NaOH(aq).
2 When a sample containing equimolar quantities of each oxide is mixed with water, the solution obtained is highly acidic.
3 Only the chlorides of **Y** and **Z** give an acidic solution with water.

Based on the statements above, identify elements **X**, **Y** and **Z**.

	X	**Y**	**Z**
A	Na	S	Mg
B	Mg	P	Si
C	Na	P	Si
D	Al	S	Mg

14 Astatine (At) is an element in Group 17. Which of the following statements is correct?

A Astatine exists as a dark brown liquid at room temperature.

B HAt(aq) is a weaker acid than HC*l*(aq).

C Astatine reacts with aqueous sodium chloride to form sodium astatide and chlorine.

D The enthalpy change of formation of hydrogen astatide is less exothermic than that of hydrogen chloride.

15 A sample of 0.300 mol of HBr was decomposed in a sealed container at temperature T. The resulting equilibrium mixture was found to contain 0.015 mol of Br_2.

$$2HBr(g) \rightleftharpoons H_2(g) + Br_2(g) \qquad \Delta H \text{ is positive}$$

Which of the following statements are true for the reaction?

1 The total number of moles of gas at equilibrium is 0.30 mol.

2 The equilibrium constant $K_c = 3.09 \times 10^{-3}$

3 The K_c will increase if reaction is carried out at higher temperature.

A 1, 2 and 3

B 1 and 2 only

C 2 and 3 only

D 1 only

16 Which statements about indicators are always correct?

1 The pH working range is greater for indicators with higher pK_a values.

2 The pK_a value of an indicator is within its pH working range.

3 The mid-point of an indicator's colour change is at pH = 7.

4 The colour red indicates an acidic solution.

A 1 and 2 only

B 2 only

C 1 and 3 only

D 3 and 4 only

17 One enzyme operates at maximum efficiency when in an aqueous solution buffered at pH 5. Which combination of substances, when dissolved in 4 dm^3 of water, would give the necessary buffer solution?

A 1 mol of HCl and 1 mol of CH_3CO_2Na

B 2 mol of NaOH and 1 mol of CH_3CO_2H

C 2 mol of NH_3 and 1 mol of $CH_3CO_2NH_4$

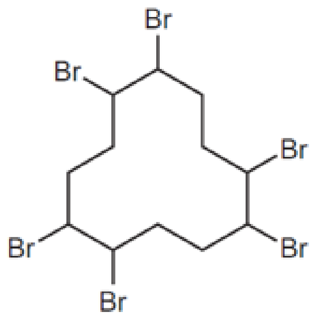
D 2 mol of CH_3CO_2H and 1 mol of NaOH

18 The alkynes are a series of hydrocarbons containing one C≡C triple bond per molecule. They have the general formula C_nH_{2n-2}.

How many structural isomers are there for the alkyne containing six carbon atom per molecule?

A 3 **B** 5 **C** 6 **D** 7

19 The compound shown is used as flame retardant.



Which statement about this molecule is **not** correct?

A The carbon atom ring is planar.

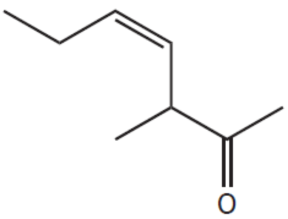
B It is immiscible in water.

C Its empirical formula is C_2H_3Br.

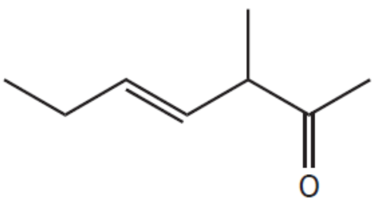
D The compound reacts with ethanolic sodium hydroxide to form $C_{12}H_{12}$.

20 5-methylhept-3-ene-2-one is an aroma molecule found in some types of tea. Which of the following shows a structural formula for the *trans*-isomer of 5-methylhept-3-ene-2-one?

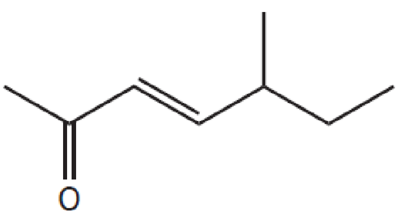
A



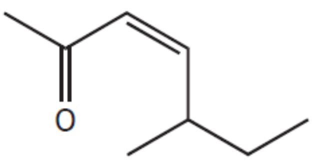
B



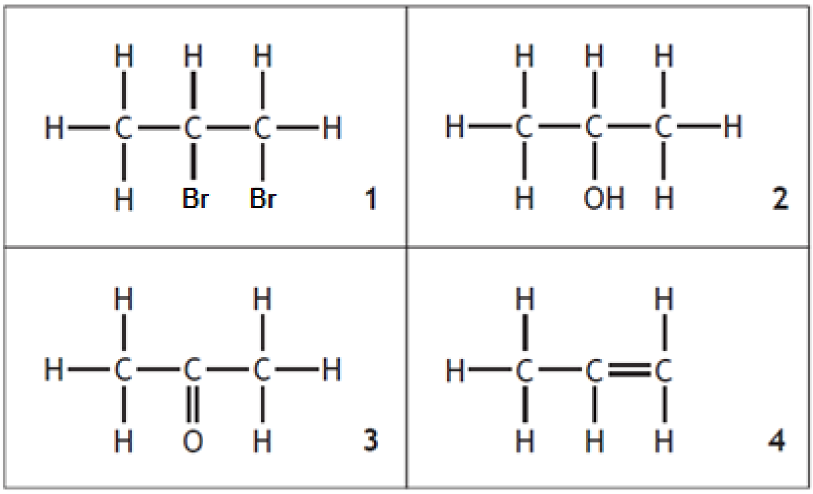
C



D



21



Which line in the table correctly identifies **W**, **X**, **Y** and **Z** in the reaction sequence?

$$W \xrightarrow{\text{reduction}} X \xrightarrow{\text{elimination}} Y \xrightarrow{\text{addition}} Z$$

	W	**X**	**Y**	**Z**
A	1	4	2	3
B	3	2	1	4
C	3	2	4	1
D	4	1	2	3

22 2-methylbuta-1,3-diene, $CH_2{=}C(CH_3){-}CH{=}CH_2$, is used as a monomer in the manufacture of synthetic rubbers.

Which compound would **not** produce this monomer on treatment with excess concentrated sulfuric acid at 170°C?

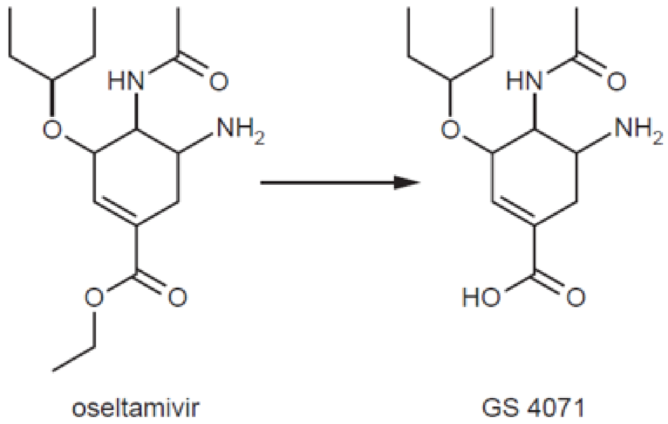
A $(CH_3)_2C(OH)CH(OH)CH_3$

B $HOCH_2CH(CH_3)CH_2CH_2OH$

C $HOCH_2CH(CH_3)CH(OH)CH_3$

D $HOCH_2C(CH_3)(OH)CH_2CH_3$

23 Oseltamivir is an anti-viral drug that is converted to its active form, GS 4071, in the body a being administered. Assume that R-O-R is an inert functional group.



Which of the following statements are correct?

1 The reaction shown above is an elimination reaction.

2 There are three single C-C σ bonds formed by sp^2 - sp^2 overlap in oseltamir and GS 4071.

3 When oseltamir is hydrolysed by aqueous HC*l*, 3 products are obtained.

4 The common functional groups found in oseltamir and GS 4071 are amine, alkene and ketone groups.

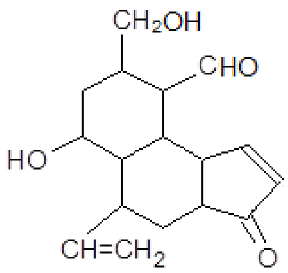
A 1, 2, 3 and 4 only

B 2, 3 and 4 only

C 1 and 4 only

D 3 only

24 What is the correct number of hydrogen atoms incorporated per molecule of compound **Y** when it is reacted with each of the following reducing agents?



Compound **Y**

	Reducing agent	Number of hydrogen atoms incorporated per molecule of **Y**
1	H_2 / Ni	8
2	$LiAlH_4$ in dry ether	8
3	$NaBH_4$ in ethanol	4

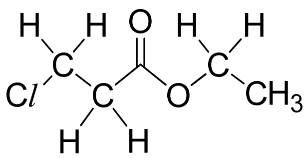
A 1 and 2 only

B 1 and 3 only

C 1 only

D 2 and 3 only

25 What would be the products formed when the following compound is boiled with aqueous sodium hydroxide?



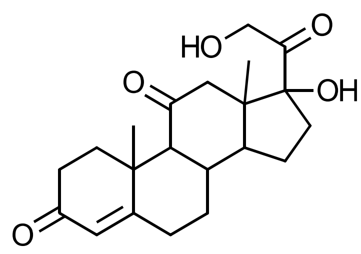
A CH_3CH_2OH and $HOCH_2CH_2CO_2^-Na^+$

B CH_3CH_2OH and $ClCH_2CH_2CO_2^-Na^+$

C $CH_3CH_2O^-Na^+$ and $ClCH_2CH_2CO_2^-Na^+$

D $CH_3CH_2O^-Na^+$ and $HOCH_2CH_2CO_2^-Na^+$

26 Cortisone is one of the main hormones that are released in response to stress.



cortisone

Which statement about this compound is **incorrect**?

A It does not exhibit cis-trans isomerism.

B It will decolourise hot acidified $Cr_2O_7^{2-}$ ions.

C It can undergo reaction with an amine in presence of dicyclohexylcarbodiimide to form an amide.

D It can undergo esterification with ethanoic acid, in the presence of H^+ ions.

27 What is the maximum size of a nanoparticle?

A 1×10^{-5} cm

B 1×10^{-7} cm

C 1×10^{-9} cm

D 1×10^{-11} cm

28 Which statement best explains that geckos are able to stick on walls?

A The millions of tiny setae on its feet can be attached on the rough surface of the wall.

B The bending of the tiny setae on the surface of the wall causes surface area of contact to increase.

C Its body weight on the wall causes a reactive force from the wall to itself which is sufficient to support its own body weight.

D The millions of tiny setae on the wall have a large surface area to volume ratio to generate extensive hydrogen bonds to support its own body weight.

29 Which of the following statements are correct for a thermosetting plastic?

1 strong covalent bonds formed between the polymer chains

2 carbon-carbon covalent bonds joining the monomers together

3 it can be recycled

4 it is rigid and has high tensile strength

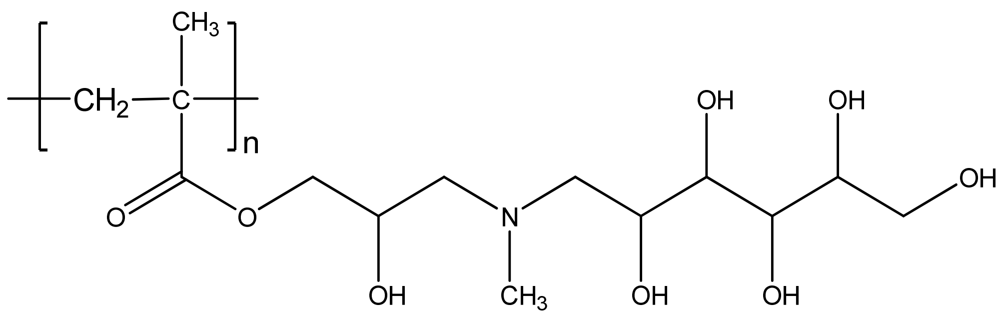
A 1 and 2 only

B 1, 3 and 4 only

C 1 and 3 only

D 1, 2 and 4 only

30 The structure below is that of a polymer based on N-methyl-D-glucamine.



Which statements correctly describe the above polymer?

1 It is able to form hydrogen bonds with water.

2 It can be classified as a condensation type of polymer.

3 The polymer has no reaction with hot aqueous base.

4 The monomer cannot exhibit cis trans isomerism.

A 1 and 2 only

B 1 and 4 only

C 2 and 4 only

D 2 and 3 only